Revision

Revision index	Entry date	Change description	Person responsible, name, signature, date	CR Number	Remark	Page
1	2	3	4	5	6	7
-	21.05.2021	New drawing	D. Ehrlich	--	-	0
a	14.10.2021	Update: starter wiring/relays KF1.0a&b/relais wiring	D. Ehrlich /M. Mayer	CR020513	CN029547	0

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designation
circuit-diagram "control-panel" BR4000ECU9 DEIF AGC-4 A10

project number
XZ54934000031 (ZNG00040840)

material number
see table-item

serial number

type-of-system
SUP42

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PROJEKT INFORMATION

Wire labelling:	No	
Protection:	IP54	
Mains connection:	Rated voltage	400V AC
	Mains frequency:	50Hz
	Wires of cable:	3 Phase / N / PE
	Cross section max.:	5x10mm²
	Max load:	22A
	Max Icu:	25kA

Part list number	remark
X54934400437	DeiF AGC 4
X54934400438	DeiF AGC 4 mit IOT

INFORMATION TO SWITCHGEAR ASSEMBLY

	generally	auxiliary-drives	Generator	voltage sensing busbar / mains
type-of-electrical-system	TN-S			
rated operating-voltage Ue:		max. 440 VAC	max. 690 VAC	max. 690 VAC
rated voltage Un:		440 VAC	690 VAC	690 VAC
rated insulation voltage Ui:		440 VAC	690 VAC	690 VAC
rated impulse withstand voltage Uimp		4 kV	6 kV	6 kV
rated current In:		**3,0 bis 25,0A**		
max. back-up-fuse:				
rated conditional short-circuit current Icc:				
rated diversity factor RDF:	**0,08 - 0,68**			
rated frequency fn:	50 Hz			
overvoltage-category:	III			
pollution-degree:	3			
EMC ambience:	A			
protection-class:	IP 54			
min. ambient-temperature:	0 °C			
max. ambient-temperature:	45 °C			
max. installation-altitude:	2000 m			
auxiliary-voltage DC:	24 V			
auxiliary-voltage AC:	24 V			

Type label:

Client:	# # #
Series no.:	# # #
Type designation:	MIP BR4000
Material no.:	# # #
Type of electricyl system:	TN-S
Rated Voltage Un:	415V AC
Rated Frequency fn:	50 Hz
Rated current InA:	# # #
Rated short current ICC:	25 kA
Control Voltage:	24V DC
Protection class:	IP 54
Year of manufacturing:	# # #
Listed standards:	IEC 61439-1 IEC 61439-2 DIN EN 60204-1

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			creator	E56395	17.05.2021	customer-description	project-description
			editor.	Dieter Ehrlich	21.10.2021		circuit-diagram "control-panel" BR4000ECU9 DEIF AGC-4 A10
			checked	F.Jäger	12.02.2020		
revision	date	name	norm	M.Prinz	12.02.2020		

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Page Description: Table of contents	drawing-number: BR2000-4000ECU9	order number	EPLAN P8: 2.8.3	== + &EAB =
	material number: see table-item	status: with Connectivity Box	format: A3	version: Standard; page (&EAB) 5 from 10; Full pages 6 from 87

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TECHNICAL DATA AND CONSTRUCTION DETAILS

IEC 61355

Full designation	Structure description
mounting-location	
+BSF	Baseframe
+ENG	Engine
+GEN	Generator incl. terminal box
+MIP	MTU Genset interface panel
+MMC	MTU Genset control
+GCB	Generator circuit breaker
+MCB	Mains circuit breaker
+FAN	Fan control
+TNK	Fuel tank
+EXT	Devices and enclosures outside any other enclosure, not delivered by MTU
+CUS	Customer page
+DOK	documents

Full designation	Structure description
document-type	
&EAA	Electrical cover sheet
&EAB	Electrical table of contents
&EDB	Description
&EFS	Wiring diagram
&ETL	Layout
&EMA	Electrical terminal diagram
&EMB	Electrical cable plan
&EPC	Bill of material

Full designation	Structure description
system	
=EKW	Electrical power
=CON	Control voltages
=EES	Engine emergency stop
=PLC	Programable logic controller
=COM	Communication
=STR	Engine starting
=MCC	Mixture cooling circuit
=ECC	Engine/jacket water circuit
=HWC	Heating water circuit
=FUL	Fuel supply

TECHNICAL DATA AND CONSTRUCTION DETAILS

line and color definition for potential tracing

(Colors are only visible if you use colored print out)

Supply, Line and measure voltage	L1, L2, L3 (120V, 230V, 400V ,690V, ...)	L
Supply, Line and measure voltage	N (120V, 230V, 400V ,690V, ...)	N
Protective earth	Ground	PE
Control voltage + 24VDC, 12VDC L+	L+	L+
Control voltage - 0VDC L-	L-	L-
Shield	SH	shield
Starter battery cabeling +24V	L-	L+Batt
Starter battery cabeling -24V	L-	L-Batt

line and color definition for potential tracing

(Colors are only visible if you use colored print out)

Control voltage 24VAC 24VAC L		24VAC-L
Control voltage 24VAC 24VAC N		24VAC-N
External source voltage	....	ext
Analog signals	QSOLL+, QSOLL-	analog
Current transformer	L1-l-Geno, L1-k-Geno	L1-k-Geno
Emergency stop	L+	NA
All other connections EPLAN standard		

TECHNICAL DATA AND CONSTRUCTION DETAILS

Wire colors within enclosure

According
IEC EN 60204-1

Rated voltage	Above 48VAC	black
Neutral wire	N	light blue
Protective earth	PE	green/yellow
Control voltage	24-240V AC	red
	0V AC	red/white
	24-110V DC	dark blue
	0V DC	grey
PLC-Controller		white
Measuring technique		grey
wires before Mainswitch		orange

Main power before main switch	L1 - L2 - L3
Main power main switch on	1L1 - 1L2 - 1L3
Neutral wire	N
Protective conductor	PE

Mains power supply	400VAC	-XD00
Power supply Heating GEN	230VAC	-XD01
Power supply heatg. cool. water	400VAC	-XD02
Emergency stop		-XD06
Supply DC	24VDC	-XDL+
Supply DC	0VDC	-XDL-
Harness W011-ECU9 Digital		-1XD01
Harness W011-ECU9 Analog		-1XD02
Harness W020 Engine		-2XD01
Harness W030 Generator signal		-3XD01/-3XD02 -3XD03
Harness W031 Generator PT100		
Option		
Customer interface WD04.1 / WG04.2 / WG04.3		-4XD01/-4XD02/ -4XD03
Harness W050 Power panel		-5XD01
Control cable WD06.1 Generator circuit breaker		-6XD01
Control cable WD07.1 Mains circuit breaker		-7XD01
Control cable WD08.1 Fan control enclosure		-8XD01
Control cable WD09.1/WD09.2 Fuel control		-9XD01/-9XD02
CAN bus		-XF01

Tightening torque

Power relay	75A	M3,5x5 M5x8	1,1-1,2 Nm 3,2-3,5 Nm
Starter	Prestolite 9KW	M5 M12	2,0-3,0 Nm 27-33 Nm
Starter	Prestolite 15KW	M5 M10	1,5-1,7 Nm 9-10 Nm
Starter	Delco 9KW	Housing 10-32 INCH 1/2-13 INCH	18-23 Nm 1,8-3,4 Nm 27-34 Nm
Battery disconnect switches		M10 M12	15-20 Nm 18-22 Nm
Battery terminals	DIN 72331	Cable M6 M8	5-6 Nm Class 4,6 / 3,5 Nm Class 4,6 / 8,4 Nm
Grounding		M8 M10 M12 M14	Class 8,8 / 28 Nm Class 8,8 / 54 Nm Class 8,8 / 93 Nm Class 8,8 / 148 Nm